NEXUS-1

FUNCTIONAL GUIDE

Nuclear Energy Facility — Operational Intelligence Console

A screen-by-screen walkthrough, organised by the application menu

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DEMONSTRATION SOFTWARE

Nothing here is connected to a real plant. Models are studied and representative, not validated. The platform has no control authority.

About this guide

NEXUS-1 is a single-file demonstration console that simulates the monitoring, analytics and decision-support layers of a nuclear energy facility. This guide describes what every screen does and how to use it. It follows the application's left-hand **navigation menu** from top to bottom: each menu group becomes a numbered section, and each screen within it is described in turn.

The console behaves as a connected system rather than a set of isolated pages. A single operator action — most notably a reactor **SCRAM** — propagates across the whole console: it appears in the incident timeline, is hash-sealed into the compliance log, increments rod-demand cycles on the inspection screens, accelerates component ageing, and nudges equipment health in the Component Registry. Where a screen feeds or is fed by another, this guide notes it under *Connected to*.

Conventions used throughout the console

Status colour	**Green** nominal · **Amber** degrading / monitor · **Red** limit reached or tripped
Unit selector	The dropdown in the top bar chooses the active reactor unit; most reactor screens follow this selection.
? button	The top-bar question mark opens context help for whatever screen you are viewing.
Fleet	Five units ship by default: two PWR units (900 / 1300 MWe) and three NUWARD-class SMR modules (170 MWe each).

Contents

Overview, Fleet & Plant 3D 1

Overview

Menu: Overview (top of menu)

The whole-plant landing screen. It presents the headline state of the selected unit and the fleet at a glance: active power, core conditions, safety margins and a plant-state indicator, alongside a compact process flow from reactor to grid that updates live.

- Read the key parameters and overall health without drilling into any subsystem.
- Switch the active unit from the top-bar selector and watch the snapshot re-point.

Connected to: the rest of the console — every detailed screen expands on something summarised here.

Plant Fleet

Menu: Plant Fleet

The fleet is modular and scalable: a mix of large PWR units and small modular reactor (SMR) modules, each synchronised to a shared 400 kV grid bus. This screen shows every unit as a card with its class, capacity, live output and status, plus a bus diagram aggregating online units into total plant output.

- Toggle any PWR unit or SMR module on or off; plant output is the sum of the online units.
- Select a unit to make it the active one that drives the detailed Reactor pages.

Connected to: Overview, Plant 3D View, Component Registry and all Reactor screens, which follow the selected / online state.

Plant 3D View

Menu: Plant 3D View

An animated three-dimensional schematic of the entire fleet in one scene: PWR containments and SMR modules feed a common steam header into a steam generator, then a turbine-generator and the grid. Active units glow and pulse, spin the turbine, and emit steam that flows visibly along the pipework; standby units are dark.

- Drag the scene to rotate; toggle auto-rotation, labels and steam flow from the bar at the top.
- Click a reactor chip to switch that unit on or off — this also changes it across the whole plant.
- Floating labels mark each reactor (with live MW), the steam generator, turbine and generator.

Connected to: Plant Fleet — it shares the same online/output state, so toggling here is equivalent to toggling on the Fleet page.

Reactor 2

Core

Menu: Reactor › Core

The reactor core layout and overall reactor state for the selected unit. A fuel-assembly map renders the power / flux distribution across the core, sized to the unit type (a full PWR lattice or a smaller SMR core).

- Inspect the fuel-assembly map and power distribution.
- Read overall reactivity and the prevailing safety margins.

Control Rods

Menu: Reactor › Control Rods

Interactive control-rod operation for the selected unit. Moving the rod bank changes reactivity, and power, neutron flux and grid output respond through the live physics. A SCRAM fully inserts the rods within seconds.

- Drag the rod-bank slider, withdraw to a target, or trigger a SCRAM and watch the plant respond.
- A SCRAM is a consequential action — it is logged and counted across the console (see below).

Connected to: Incident Analysis and Compliance / Audit (a SCRAM is recorded automatically), and Rod Inspection, Aging and the Component Registry, which all treat SCRAM cycles as wear.

Neutronics

Menu: Reactor › Neutronics

Neutron-flux and reactivity detail for the selected unit: the flux distribution and a breakdown of the components that sum to total reactivity.

- Read the flux distribution and the individual reactivity contributions.

Reactor Kinetics

Menu: Reactor › Reactor Kinetics

A physically real point-kinetics simulator — the analytical heart of the console. It integrates the point-kinetics equations with six delayed-neutron groups and couples them to Doppler (fuel-temperature), moderator-temperature and xenon-135 feedbacks, using a lumped two-node thermal model and a stable operator-splitting integration scheme.

- Apply a small positive reactivity step (in pcm) and watch the prompt jump followed by self-regulation, then SCRAM.
- After a shutdown, switch to accelerated time (e.g. 600x) to watch the xenon transient build and decay.

Connected to: the Control Rods and 3D Reactor View, which drive and visualise the same dynamics.

3D Reactor View

Menu: Reactor › 3D Reactor View

Two side-by-side three-dimensional cores forming a digital twin of the selected unit: the left core is the unit as seen through (simulated) sensors — noisy and slightly lagged — while the right core is its clean expected model. The same controls drive both.

- Drag a core to rotate it; fuel colour encodes temperature.
- Sustained divergence between the measured and modelled core is the digital-twin early-warning signal.

Connected to: Digital Twin, which quantifies the same model-versus-telemetry divergence.

Steam & Secondary

Menu: Reactor › Steam & Secondary

The secondary side for the selected unit: steam generators, the turbine train (HP and LP turbines, moisture-separator reheat) and condenser, with live steam pressure, flow and feedwater readings that track core power.

- Follow the heat path from the steam generators through the turbine train to the condenser.
- Watch secondary parameters respond as reactor power changes.

Coolant / TH

Menu: Reactor › Coolant / TH

Primary coolant and thermal-hydraulic conditions: loop flows, hot- and cold-leg temperatures, pressuriser level and pressure, and feedwater — the conditions that carry heat away from the core.

- Monitor primary loop flows and temperatures and the pressuriser state.
- See how thermal-hydraulic conditions follow reactor power and rod position.

Rod Inspection 3

Inspection Overview

Menu: Rod Inspection › Inspection Overview

A fleet-level quality dashboard for control-rod and related rod assemblies. It summarises the inspection status across rod types, flagging which populations have accepted, rework or rejected items.

- Read the overall rod-quality picture before opening an individual rod type.

Rod Types (per type)

Menu: Rod Inspection › (each rod type)

Each rod type — control, safety / shutdown, burnable-poison, instrumentation and source rods — has its own non-destructive testing (NDT) screen. It generates a simulated radiograph of a deterministic rod inventory (serial numbers, materials, cladding, defects) and applies the six standard NDT methods to reach an accept / rework / reject verdict per rod.

- Inspect a simulated radiograph and the detectable defects for the selected rod type.
- Safety and control rods display the SCRAM demand cycles accumulated during this session.

Connected to: Control Rods — SCRAM activity from this session feeds the rod-demand counts shown here.

NDT Methods

Menu: Rod Inspection › NDT Methods

A reference screen describing the six non-destructive testing methods used across the inspection module — radiography, ultrasonic, visual, eddy-current, dye-penetrant and magnetic-particle testing — and the defect classes each is able to detect.

- Use as a key for interpreting the verdicts shown on the per-type inspection screens.

Personnel 4

Overview

Menu: Personnel › Overview

Staffing and shift-coverage for the plant, organised by sector (operations, nuclear safety, maintenance, radiation protection, chemistry and others). It shows required versus present roles and highlights minimum-staffing gaps.

- See current coverage by sector and whether minimum-staffing rules are met.

Sectors (per sector)

Menu: Personnel › (each sector)

Each staffing sector has its own roster screen listing roles, named staff and licence categories, with the critical roles that must be filled for the plant to operate.

- Drill into a sector to see its roster and which roles are safety-critical.

Stress Test

Menu: Personnel › Stress Test

An absence-resilience test: remove staff and see whether each sector can still meet its minimum-staffing and critical-role requirements.

- Simulate absences and identify which sectors lose coverage first.

Plant Lifecycle 5

Aging & Degradation

Menu: Plant Lifecycle › Aging & Degradation

Long-term life management for the selected unit. A handful of major, mostly passive components (reactor pressure vessel, primary piping, steam generators, containment concrete, cables and core internals) are tracked for life consumed against their limiting mechanisms, each with a trend sparkline.

- See life consumed per component and the dominant degradation mechanism.
- Ageing grows with simulated operating time and accelerates with SCRAM thermal cycles.

Connected to: Control Rods (SCRAM cycles) and the Component Registry, which is the operational counterpart to this long-term view.

Decommissioning

Menu: Plant Lifecycle › Decommissioning

The cradle-to-grave lifecycle. A unit can be moved into decommissioning, after which the screen walks the dismantling stages, the radiological inventory decaying over time, and the accumulating waste volumes by classification.

- Begin decommissioning a unit (it is shut down first) and follow the staged programme.
- Watch radiological activity decay and very-low / low / intermediate-level waste volumes accumulate.

Connected to: Plant Fleet — decommissioning a unit takes it offline across the console.

Robotics & Vehicles 6

Fleet Overview

Menu: Robotics & Vehicles › Fleet Overview

Status of the robotic and vehicle fleet used for inspection, survey, manipulation, decontamination and emergency tasks — each asset with its availability, battery charge and accumulated radiation dose.

- See which robots are available and their charge / dose budget at a glance.

Categories (per category)

Menu: Robotics & Vehicles › (each category)

Each robot category lists its individual assets with per-asset status, battery level and cumulative radiation dose (robot electronics have a finite radiation tolerance).

- Inspect the assets within a category and their remaining dose budget.

Mission Readiness

Menu: Robotics & Vehicles › Mission Readiness

A capability test: each standard mission needs a capable robot that is available, sufficiently charged and still has dose budget remaining. Stress the fleet and see which missions can still be covered.

- Identify which missions remain coverable as the fleet is stressed.

Zone Access 7

Live Presence

Menu: Zone Access › Live Presence

Live occupancy and access state across the plant's controlled zones, with any access violations highlighted.

- Monitor who is present in which zone and flag violations.

Permissions Matrix

Menu: Zone Access › Permissions Matrix

The zone-authorisation policy: which roles are permitted in which zones, presented as a matrix.

- Read the access policy that the Live Presence screen enforces against.

Power & Grid 8

Power & Grid

Menu: Power & Grid

The turbine-generator and grid interface: active and reactive power, generator voltage and power factor, turbine speed, grid frequency and synchronisation state on the 400 kV bus.

- Monitor electrical output and grid synchronisation as plant power changes.

Radiation / Safety 9

Radiation / Safety

Menu: Radiation / Safety

Radiation and dose monitoring: area readings measured against limits, with the standing reminder that reactor protection systems are hardware-isolated, deterministic and human-supervised.

- Read area dose rates against their limits.

Alarms & Events 10

Alarms & Events

Menu: Alarms & Events

Active alarm management: a live, prioritised list of plant alarms and events that can be acknowledged and tracked. Alarms are raised both by the simulation and by other modules — for example a degrading component in the registry or a manual SCRAM.

- Acknowledge alarms and track their state.
- The sidebar badge shows the current active-alarm count.

Connected to: Component Registry and Control Rods, which raise alarms here automatically.

AI Diagnostics 11

AI Diagnostics

Menu: AI Diagnostics

Supervised predictive diagnostics, presented as advisory only. It surfaces failure-probability and degradation forecasts for plant equipment — for example the likelihood of coolant-pump degradation over a coming window.

- Read failure-probability and degradation forecasts.
- This is a supervised predictor, distinct from the RL optimiser which learns control policies.

Connected to: Component Registry, whose live equipment health and predictive digest feed this view.

Optimization (RL) 12

Advanced Optimization (RL)

Menu: Optimization (RL) — BETA

An experimental reinforcement-learning optimiser, clearly marked BETA and advisory. The agent is trained entirely inside the digital twin, never on a live plant, and the learned policy is exposed as a readable table rather than an opaque model.

- Train the agent in the twin and watch the reward curve climb.
- Inspect the learned policy as a transparent table.
- Advisory only, with safety clamps — no control authority on a real plant.

Connected to: Digital Twin, which is the sandbox the agent trains inside.

System Dependencies 13

System Dependencies

Menu: System Dependencies

A model of how plant subsystems influence one another, in three views. A dependency graph lets you click any node to trace everything it affects downstream with approximate propagation delays; an influence matrix gives the weighted coupling between any two components; and a causal-chain view reconstructs a reactor trip with a what-if toggle. Node colours are driven live by the selected unit.

- Graph: click a node to highlight its downstream influence and delays.
- Matrix: click a cell to read the influence weight of one component on another.
- Causal Chain: step through the reactor-trip reconstruction and toggle a what-if.

Connected to: Component Registry, where each tracked component maps onto a node in this graph.

Component Registry 14

Component Registry

Menu: Component Registry

A per-unit inventory of plant equipment with live health. Every component (pumps, steam generators, the vessel, rod-drive mechanisms, turbine, generator, diesels, containment and more) carries a health value that responds to operation: running at power wears rotating equipment and every SCRAM cycle ages the coolant pumps and rod drives. PWR units list the loop-type plant; SMR modules list the integral configuration (in-vessel steam generator, compact canned pumps, a passive heat sink). A predictive digest highlights the highest-risk item with an estimated time to the maintenance limit and a degradation probability.

- Browse equipment grouped by system, each with health, status, redundancy and an estimated maintenance horizon.
- See the part-set differ between PWR (loop-type) and SMR (integral) units.
- When a component degrades or faults, an alarm is raised automatically.

Connected to: Alarms (faults raise alarms), AI Diagnostics (the digest feeds it), System Dependencies (each item maps to a node), and Aging — the long-term life view of the same plant.

Digital Twin 15

Digital Twin

Menu: Digital Twin

Model-versus-reality reconciliation: it compares the model's expectation against the (simulated) telemetry stream and reports the divergence, which under real conditions would be the early indicator of a sensor fault or model error.

- Read the running comparison between modelled and measured plant state.

Connected to: 3D Reactor View, which renders the same telemetry-versus-model idea spatially.

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Trends & History

Menu: Trends & History

Historical time-series of the key plant parameters, so behaviour can be tracked over time rather than only as an instantaneous snapshot.

- Track selected parameters over a rolling history window.

Incident Analysis 17

Incident Analysis

Menu: Incident Analysis

An event timeline. Live SCRAM events from this session appear at the top automatically as they happen, above a baseline event reconstruction that illustrates a worked incident.

- See SCRAMs you trigger appear live at the top of the timeline.
- Review the baseline reconstructed event below.

Connected to: Control Rods — every SCRAM writes a live entry here.

Root Cause Graph 18

Root Cause Graph

Menu: Root Cause Graph

Turns an alarm flood into a single causal story. An engineered fault-propagation backbone (derived from the plant fault trees) is lit live by active alarms; you can test any node as the root cause to see which alarms it explains. An alarm-flood demonstration collapses around fourteen cascading alarms down to one origin plus its downstream cascade, and a ranking view shows how causal-origin reasoning points upstream to the true cause where naive alarm-counting points only at the most-alarmed symptom.

- Click any node to test it as the root cause and see the alarms it accounts for.
- Run the alarm-flood demonstration and watch the cascade collapse to a single origin.
- Advisory only — it informs operators and investigators and never feeds control or safety actuation.

Connected to: Alarms and System Dependencies, whose fault relationships it reasons over.

Compliance / Audit 19

Compliance / Audit

Menu: Compliance / Audit

A tamper-evident action log: a hash-sealed record of operator and system actions providing complete traceability. Safety-significant events such as SCRAMs are written here automatically.

- Review the immutable, hash-sealed log of actions and events.
- SCRAMs and other safety actions are recorded as Safety-type entries automatically.

Connected to: Control Rods and Incident Analysis, which generate the entries recorded here.

Security / Services 20

OT Security

Menu: Security / Services

The operational-technology security posture of the platform: zero-trust architecture, network segmentation, hardware security modules, multi-factor authentication and intrusion-detection status — reflecting the critical-infrastructure requirements a real deployment would carry.

- Review the OT security controls and their current status.

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Help & Guide

Menu: Help & Guide

The in-application guide. It organises every screen by theme and explains how to read the console; the top-bar ? button gives the same help for whichever screen you are currently viewing.

- Browse all screens by theme, or press ? for help on the current screen.

About

Menu: About

Author and scope. This screen states plainly what is real versus simulated in NEXUS-1 and gives the author's background — the right place to start for understanding the project's intent.

- Read the project's scope, what is and is not real, and the author background.

Scope, Honesty & Authorship

NEXUS-1 is a portfolio demonstration of software architecture, simulation and visualisation applied to the nuclear energy domain. It is deliberately explicit about its limits:

- Nothing in the console is connected to a real plant. All telemetry is generated by the built-in simulation.
- The physics — most fully the point-kinetics engine — is studied and representative, not a validated or licensed model.
- Predictive and optimisation features (AI Diagnostics, the RL optimiser, the Component Registry digest) are advisory and, where accelerated or illustrative, are labelled as such. They are not identified from plant data.
- The platform has no control authority. In a real architecture, certified reactor-protection systems remain hardware-isolated, deterministic and human-supervised; this console would sit only in the monitoring, analytics and decision-support layers.

This document and the NEXUS-1 console are the original work of Grigorios Agathangelidis. Build signature NX1-094C79E0A366D34F is embedded throughout the application as an authorship mark.